# Lifetime risk of developing impaired glucose metabolism and eventual progression from prediabetes to type 2 diabetes: a prospective cohort study

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## Summary

**Background** Data are scarce for the lifetime risk of developing impaired glucose metabolism, including prediabetes, as are data for the risk of eventual progression from prediabetes to diabetes and for initiation of insulin treatment in previously untreated patients with diabetes. We aimed to calculate the lifetime risk of the full range of glucose impairments, from normoglycaemia to prediabetes, type 2 diabetes, and eventual insulin use.

**Methods** In this prospective population-based cohort analysis, we used data from the population-based Rotterdam Study. We identified diagnostic events by use of general practitioners' records, hospital discharge letters, pharmacy dispensing data, and serum fasting glucose measurements taken at the study centre (Rotterdam, Netherlands) visits. Normoglycaemia, prediabetes, and diabetes were defined on the basis of WHO criteria for fasting glucose (normoglycaemia: ≤6·0 mmol/L; prediabetes: >6·0 mmol/L and <7·0 mmol/L; and diabetes ≥7·0 mmol/L or use of glucose-lowering drug). We calculated lifetime risk using a modified version of survival analysis adjusted for the competing risk of death. We also estimated the lifetime risk of progression from prediabetes to overt diabetes and from diabetes free of insulin treatment to insulin use. Additionally, we calculated years lived with healthy glucose metabolism.

**Findings** We used data from 10 050 participants from the Rotterdam Study. During a follow-up of up to 14·7 years (between April 1, 1997, and Jan 1, 2012), 1148 participants developed prediabetes, 828 developed diabetes, and 237 started insulin treatment. At age 45 years, the remaining lifetime risk was 48·7% (95% CI 46·2–51·3) for prediabetes, 31·3% (29·3–33·3) for diabetes, and 9·1% (7·8–10·3) for insulin use. In individuals aged 45 years, the lifetime risk to progress from prediabetes to diabetes was 74·0% (95% CI 67·6–80·5), and 49·1% (38·2–60·0) of the individuals with overt diabetes at this age started insulin treatment. The lifetime risks attenuated with advancing age, but increased with increasing BMI and waist circumference. On average, individuals with severe obesity lived 10 fewer years without glucose impairment compared with normal-weight individuals.

**Interpretation** Impaired glucose metabolism is a substantial burden on population health, and our findings emphasise the need for more effective prevention strategies, which should be implemented as soon in a person's life as possible. The substantial lifetime risk of prediabetes and diabetes in lean individuals also supports risk factor control in non-obese individuals.

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## Introduction

People with blood glucose concentrations that are raised but below the threshold for type 2 diabetes (ie, people with prediabetes) have an excess risk of diabetes.[1–3] Today, more than 382 million people live with diabetes worldwide and, because of the increasing prevalence of prediabetes and the rapid conversion of prediabetes to type 2 diabetes, the number is predicted to exceed half a billion by 2035.[4] Moreover, many patients with diabetes are unable to achieve glycaemic control goals through diet and oral drugs alone, ultimately requiring insulin treatment.[5–7] Previous estimates of the progression from prediabetes to diabetes and ultimately to the need for insulin treatment are scarce and have been limited to annual incidences and absolute risks within a short time period.[2]

Lifetime risk provides an estimate of the cumulative risk of developing a disease during an individual's remaining lifespan, which is relevant for patients, clinicians, and health-care policy makers.[8–10] A few studies have provided estimates of the lifetime risk of type 2 diabetes in the USA and Australia.[10–12] However, lifetime risk estimates based on accurate and careful documentation of raised blood glucose concentrations, diabetes diagnosis, and diabetes drug use are scarce. Prospective population-based cohort studies with

## Research in context

### Evidence before this study

Lifetime risk estimates provide the cumulative risk of developing a disease during an individual's remaining lifespan. Previously, lifetime risk estimates have only been reported for overt type 2 diabetes on the basis of questionnaire data and not directly from a population with a long follow-up. Progression rates from prediabetes to diabetes have mostly been estimated for 10 years and not for the rest of an individual's remaining lifespan. So far, no previous study has included the direct calculation of the lifetime risks of developing the full range of impaired states of glucose metabolism by use of empirical data from a prospective population-based cohort study.

### Added value of this study

Our findings showed that, in a well defined European population with long-term follow-up, half the normoglycaemic population eventually developed prediabetes, and most individuals with prediabetes eventually progress to overt type 2 diabetes. Furthermore, the lifetime risk to use insulin treatment for an individual aged 45 years not using insulin was 9%, suggesting that a high proportion of our population eventually uses insulin. Obesity substantially increased the lifetime risk of developing diabetes and attenuated the number of life-years lived with a healthy glucose metabolism. However, lean individuals also had a substantial lifetime risk of diabetes.

### Implications of all the available evidence

The burden that impaired glucose metabolism exerts on the population is substantial and effective prevention strategies should be implemented as soon in a person's life as possible. The substantial lifetime risks of prediabetes and diabetes in lean individuals also support the implementation of risk factor control in non-obese individuals. Furthermore, the large proportion of individuals that will eventually use insulin treatments reflects the high health-care burden of type 2 diabetes.

long-term follow-up and detailed data for the full range of impaired glucose metabolism including prediabetes, diabetes, and the eventual need for insulin treatment, allow the estimation of the burden of increased blood glucose concentrations in the context of overall survival.

In this long-term follow-up study, we used mortality rates and incidences of disease during every year of life, taking into account the competing risk of death, to assess the lifetime risk of prediabetes, diabetes, and insulin use in a large prospective population-based cohort study of individuals aged 45 years and older. Additionally, we estimated the lifetime risk of individuals with prediabetes eventually developing diabetes and for patients with diabetes ultimately progressing to insulin use.

## Methods

### Study design and population

This study is embedded within the framework of the Rotterdam Study, a prospective cohort study of the community-dwelling population aged 45 years and older in Rotterdam, Netherlands. The study design of the Rotterdam Study has been described in detail previously.[13] Briefly, in 1990 all inhabitants of a well defined district of Rotterdam were invited to participate, of whom 7983 (78%) agreed to participate. The study was extended in 2000, with a second cohort of individuals who had reached the age of 55 years or moved into the study area after 1990 (n=3011). In 2006, a third cohort was enrolled including inhabitants aged 45 years and older (n=3932), bringing the total study population to 14926 individuals. There were no eligibility criteria to enter the Rotterdam Study apart from the minimum age and residential area based on postal codes. We used the third centre visit (1997–99, n=4216) of the first cohort and the first visits of the second and third cohorts as baseline (2000–01 and 2006–08, respectively) for our analysis. We excluded 1367 individuals for whom a valid baseline fasting glucose measurement was not available and 314 individuals for whom informed consent was not obtained. Next, for the calculation of the lifetime risk of prediabetes, we included only individuals who were normoglycaemic at study baseline (n=7462). To calculate the lifetime risk of diabetes, we included only individuals who did not have diabetes at study baseline (n=8844). To calculate the lifetime risk of insulin use, we included only individuals who were free of insulin use at study baseline (n=9887). Figure 1 shows the selection of individuals for the analyses. The individuals with prediabetes (n=1382) were used to study progression from prediabetes to diabetes, and individuals with diabetes without insulin treatment (n=1043) were used to study the progression from diabetes to insulin use.

The Rotterdam Study has been approved by a medical ethics committee, in accordance with the Population Screening Act: Rotterdam Study, executed by the Netherlands Ministry of Health, Welfare and Sports. All participants in the present analysis provided written informed consent to participate and for information to be obtained from their treating physicians.

### Ascertainment of prediabetes and type 2 diabetes

Participants were followed up from the date of baseline centre visit onwards. At baseline and during follow-up, cases of prediabetes and type 2 diabetes were ascertained by use of general practitioners' records (including laboratory glucose measurements), hospital discharge letters, and serum glucose measurements from Rotterdam Study visits, which take place roughly every 4 years.[14] Diabetes, prediabetes, and normoglycaemia were defined on the basis of WHO guidelines.[15] Normoglycaemia was defined as a fasting blood glucose concentration of 6·0 mmol/L or lower; prediabetes was

defined as a fasting blood glucose concentration greater than 6·0 mmol/L and lower than 7·0 mmol/L, or a non-fasting blood glucose concentration higher than 7·7 mmol/L and lower than 11·1 mmol/L (when fasting samples were unavailable); type 2 diabetes was defined as a fasting blood glucose concentration of 7·0 mmol/L or higher, a non-fasting blood glucose concentration of 11·1 mmol/L or higher (when fasting samples were unavailable), or the use of blood glucose-lowering drugs. Information about the use of blood glucose-lowering drugs was obtained from both structured home interviews and pharmacy dispensing records.[14] At baseline, more than 95% of the Rotterdam Study population was covered by the pharmacies in the study area. All potential events of prediabetes and type 2 diabetes were independently adjudicated by two study physicians. In case of disagreement, consensus was sought from an endocrinologist. Completeness of follow-up until Jan 1, 2012, was 99·3%.

### Statistical analysis

We compared the baseline characteristics of normoglycaemic individuals, individuals with prediabetes, and individuals with type 2 diabetes using linear regression models, Kruskal-Wallis tests for continuous data, and $\chi^2$ tests for categorical data.

We calculated remaining lifetime risks at different ages for prediabetes, diabetes, and insulin use. We used a modified version of survival analysis to take the competing event of death into account for the calculation of the absolute lifetime risk (appendix p 3). Lifetime risk estimates were calculated at index ages 45, 55, 65, 75, and 85 years for men and women, both combined and separately. The lifetime risk estimates reflect the remaining risk at the index age to the age of last observation (maximum age of 107 years in our study). Additionally, to compare lifetime risks with absolute risks in a shorter time period, we calculated 10-year risks for prediabetes, diabetes, and insulin use at all index ages.

Next, we calculated the lifetime risk of diabetes in individuals with prediabetes to obtain an estimate of the lifetime risk of progressing from prediabetes to diabetes. Similarly, we calculated the lifetime risk of insulin use in individuals with diabetes not using insulin treatment at baseline.

To analyse the effect of anthropometric measures on the lifetime risk of prediabetes, diabetes, and insulin use, we computed lifetime risks at the age of 45 years stratified by BMI and waist circumference. Participants were stratified into four categories of BMI (<25 kg/m², ≥25 to <30 kg/m², ≥30 to <35 kg/m², and ≥35 kg/m²) and three categories of waist circumference based on the WHO classification scheme (for men: <94 cm, ≥94 to >101 cm, and ≥102 cm; for women: <80 cm, ≥80 to >87 cm, and ≥88 cm).[16]

To study the delay in onset of prediabetes, diabetes, and insulin use, we calculated the difference in mean disease-free survival by BMI and waist circumference strata. Because censoring precludes estimation of the mean survival time, we used Irwin's restricted mean survival to calculate the mean disease-free survival and overall mean survival.[17] Irwin's restricted mean survival is defined as the mean of the survival time up to a specific point in time, measured as the area under the survival curve up to that timepoint. Because little data from individuals aged older than 100 years were available, we set the restriction timepoint to age 100 years.

All data were analysed with IBM SPSS Statistics (version 21.0.0.1) and R (version 2.1), with the etm and survival libraries.[18,19]

### Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report. All authors had full access to all the data in the study, and the corresponding author had final responsibility for the decision to submit for publication.

## Results

We used data from 10 050 participants of the Rotterdam Study. The mean age of the population was 65·2 years (SD 9·8) and 5685 (57%) participants were women. Of the 10 050 participants at baseline, 7462 (74%) had normoglycaemia, 1382 (14%) had prediabetes, and 1206 (12%) had diabetes (table 1). At baseline, prevalences of prediabetes and type 2 diabetes increased with advancing age in both men and women and were higher in men than in women (appendix p 7). Individuals with prediabetes and diabetes had higher BMI and more unfavourable lipid profile than did normoglycaemic individuals. Furthermore, people with diabetes had a higher prevalence of stroke and coronary heart disease, and were more likely to be smokers compared with normoglycaemic individuals.

See Online for appendix

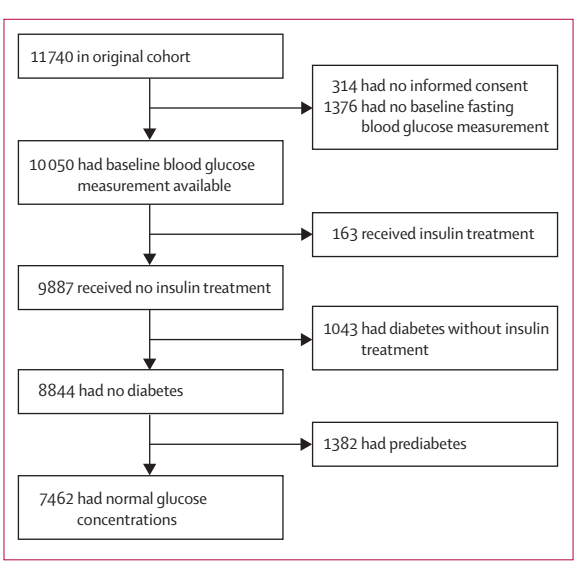


*Figure 1:* **Participant selection**

| | Participants with normal glucose (n=7462) | Participants with prediabetes (n=1382) | Participants with diabetes (n=1206) | p value |
|---|---|---|---|---|
| Women | 4411 (59%) | 692 (51%) | 582 (48%) | <0·0001 |
| Age (years) | 64·4 (9.8) | 66·6 (9·4) | 67·5 (9·6) | <0·0001 |
| White | 6948 (93%) | 1271 (92%) | 1101 (91%) | 0·04 |
| Waist circumference (cm) | 90 (11) | 96 (12 ) | 101 (12) | <0·0001 |
| BMI (kg/m²) | 26·3 (3·8) | 27·9 (4·2) | 29·4 ( 4·8) | <0·0001 |
| Total cholesterol (mmol/L) | 5·7 (1·0) | 5·8 (1·0) | 5·4 (1·1) | <0·0001 |
| HDL cholesterol (mmol/L) | 1·4 (1·2–1·7) | 1·3 (1·1–1·5) | 1·2 (1·0–1·4) | <0·0001 |
| Triglycerides (mmol/L)* | 1·3 (1·0–1·7) | 1·5 (1·1–2·1) | 1·7 (1·2–2·3) | <0·0001 |
| Non-HDL cholesterol (mmol/L) | 4·3 (1·0) | 4·4 (1·0) | 4·2 (1·1) | <0·0001 |
| LDL cholesterol (mmol/L) | 3·7 (0·9) | 3·6 (0·9) | 3·4 ( 0·9) | <0·0001 |
| Insulin (pmol/L)* | 66 (47-93) | 93 (64-133) | 110 (73-177) | <0·0001 |
| Glucose (mmol/L)* | 5·3 (5·0-5·6) | 6·3 (6·1-6·5) | 7·7 (7·0-9·5) | <0·0001 |
| eGFR (mL/min per 1·73 m²) | 81 (17) | 80 (18) | 83 (22) | <0·0001 |
| C-reactive protein (mg/L) | 1·4 (0·6-3·1) | 2·0 (0·9–4·2) | 2·5 (1·1–4·9) | <0·0001 |
| Systolic blood pressure (mm Hg) | 137 (21) | 145 (21) | 147 (22) | <0·0001 |
| Diastolic blood pressure (mm Hg) | 78 (11) | 81 (12) | 79 (12) | <0·0001 |
| Hypertension | 3448 (47%) | 873 (64%) | 875 (73%) | <0·0001 |
| History of stroke | 178 (2%) | 35 (3%) | 77 (6%) | <0·0001 |
| History of CHD | 422 (6%) | 110 (8%) | 164 (14%) | <0·0001 |
| Use of blood pressure-lowering drugs | 1408 (20%) | 437 (33%) | 474 (41%) | <0·0001 |
| Use of lipid-lowering agents | 1043 (14%) | 239 (18%) | 1178 (27%) | <0·0001 |
| Current smoking | 707 (10%) | 144 (10%) | 147 (12%) | 0·001 |
| Former smoking | 2769 (37%) | 553 (40%) | 480 (40%) | 0·001 |

Data are n (%), mean (SD), or median (IQR; for characteristics with skewed distributions). eGFR=estimated glomerular filtration rate. CHD=coronary heart disease. *Only fasting samples.

*Table 1:* **Baseline characteristics of participants, by prevalent glycaemic state**

| | n | Lifetime risk prediabetes (95% CI) | n | Lifetime risk diabetes (95% CI) | n | Lifetime risk insulin use (95% CI) |
|---|---|---|---|---|---|---|
| **45 years** | | | | | | |
| Lifetime | 7462 | 48·7% (46·2–51·3) | 8844 | 31·3% (29·3–33·3) | 9887 | 9·1% (7·8–10·3) |
| 10-year | 1233 | 8·4% (5·4–11·4) | 1344 | 3·4% (2·1–4·8) | 1400 | 0·5% (0·0–1·0) |
| **55 years** | | | | | | |
| Lifetime | 6939 | 44·5% (42·5–46·6) | 8291 | 29·2% (2·.4–31·0) | 9329 | 8·8% (7·6–10·0) |
| 10-year | 3788 | 13·2% (11·4–15·0) | 4456 | 7·0% (5·8–8·3) | 4914 | 2·2% (1·4–3·0) |
| **65 years** | | | | | | |
| Lifetime | 5109 | 37·6% (35·6–39·5) | 6257 | 24·8% (2·.1–26·5) | 7179 | 7·0% (6·0–7·9) |
| 10-year | 3901 | 19·3% (17·7–20·9) | 4754 | 11·2% (10·0–12·3) | 5414 | 3·0% (2·4–3·6) |
| **75 years** | | | | | | |
| Lifetime | 3073 | 25·8% (23·7–28·0) | 3850 | 17·4% (15·7–19·1) | 4547 | 4·7% (3·8–5·6) |
| 10-year | 2885 | 19·1% (17·3–20·9) | 3618 | 12·2% (10·9–13·6) | 4273 | 2·6% (2·0–3·2) |
| **85 years** | | | | | | |
| Lifetime | 1072 | 13·1% (10·4–15·7) | 1405 | 8·9% (6·9–10·9) | 1725 | 3·3% (2·2–4·4) |
| 10-year | 1060 | 11·9% (9·5–14·4) | 1390 | 8·2% (6·4–10·0) | 1707 | 3·3% (2·2–4·4) |

Lifetime risk of prediabetes is calculated for individuals with normal glucose levels at the index age. Lifetime risk of diabetes is calculated for individuals without diabetes at the index age. Lifetime risk of insulin use is calculated for individuals not using insulin at the index age.

*Table 2:* **Remaining lifetime and 10-year risks of prediabetes, diabetes, and insulin use, by age**

During 56 230 person-years of follow-up in normoglycaemic individuals, 1148 individuals developed prediabetes and 1343 died (prediabetes incidence rate per 1000 person-years: 20·4, 95% CI 19·3–21·6; mortality rate per 1000 person-years: 23·9, 22·7–25·2). During 69 639 person-years of follow-up in individuals who did not have diabetes at baseline, 828 individuals developed diabetes and 1709 died (diabetes incidence rate 11·9, 95% CI 11·1–12·7; mortality rate 24·5, 23·4–25·7). During 80 832 person-years of follow-up in people with type 2 diabetes who were not using insulin, 237 individuals commenced insulin use (insulin use incidence rate 3·4, 95% CI 3·0 to 3·9) and 2183 individuals died (mortality rate 27·0, 95% CI 25·9 to 28·1). The remaining lifetime risk for a 45-year-old normoglycaemic individual (non-diabetic and non-insulin user) to develop prediabetes was 48·7%, whereas the lifetime risk of diabetes and insulin use was 31·3% and 9·1%, respectively (table 2). Figure 2 shows the cumulative incidence functions in individuals aged 45 years. Lifetime risks of prediabetes, diabetes, and insulin use were attenuated with increasing age. Compared with the lifetime risks, the 10-year risks of prediabetes, diabetes, and insulin use were lower at all index ages (table 2). The remaining lifetime risks did not differ by sex, irrespective of age (appendix pp 8–9). With adjustment for the competing risk of death, the lifetime risks were lower compared with the unadjusted risk derived from Kaplan-Meier estimates (appendix p 4).

Person-years of follow-up for individuals with prediabetes was 9884 compared with 7613 person-years for individuals with diabetes. In 1382 individuals with prediabetes, 425 individuals developed diabetes and 257 individuals died without diabetes (diabetes incidence rate per 1000 person-years: 43·0, 95% CI 39·2–47·2; mortality rate per 1000 person-years: 26·0, 23·0–29·3). The lifetime risk for individuals with prediabetes aged 45 years to progress to diabetes was 74·0% (95% CI 67·6–80·5). Furthermore, of 1043 individuals with diabetes who had never been treated with insulin, 183 individuals commenced insulin use, and 302 died (insulin use incidence rate per 1000 person-years: 24·0, 95% CI 20·8–27·7; mortality rate per 1000 person-years: 39·7, 35·5–44·3). The lifetime risk for individuals with diabetes aged 45 years to start insulin treatment was 49·1% (95% CI 38·2–60·0).

Stratification by BMI showed that normoglycaemic people with healthy weight (<25 kg/m²) at age 45 years had a significantly lower prediabetes lifetime risk compared with overweight and obese individuals (table 3). Stratification by waist circumference showed similar effects on lifetime risks for diabetes in individuals with prediabetes. Similarly, in individuals with diabetes, the lifetime risk for insulin use among patients with diabetes was higher with increasing BMI and waist circumference (table 4). Figure 2 shows the

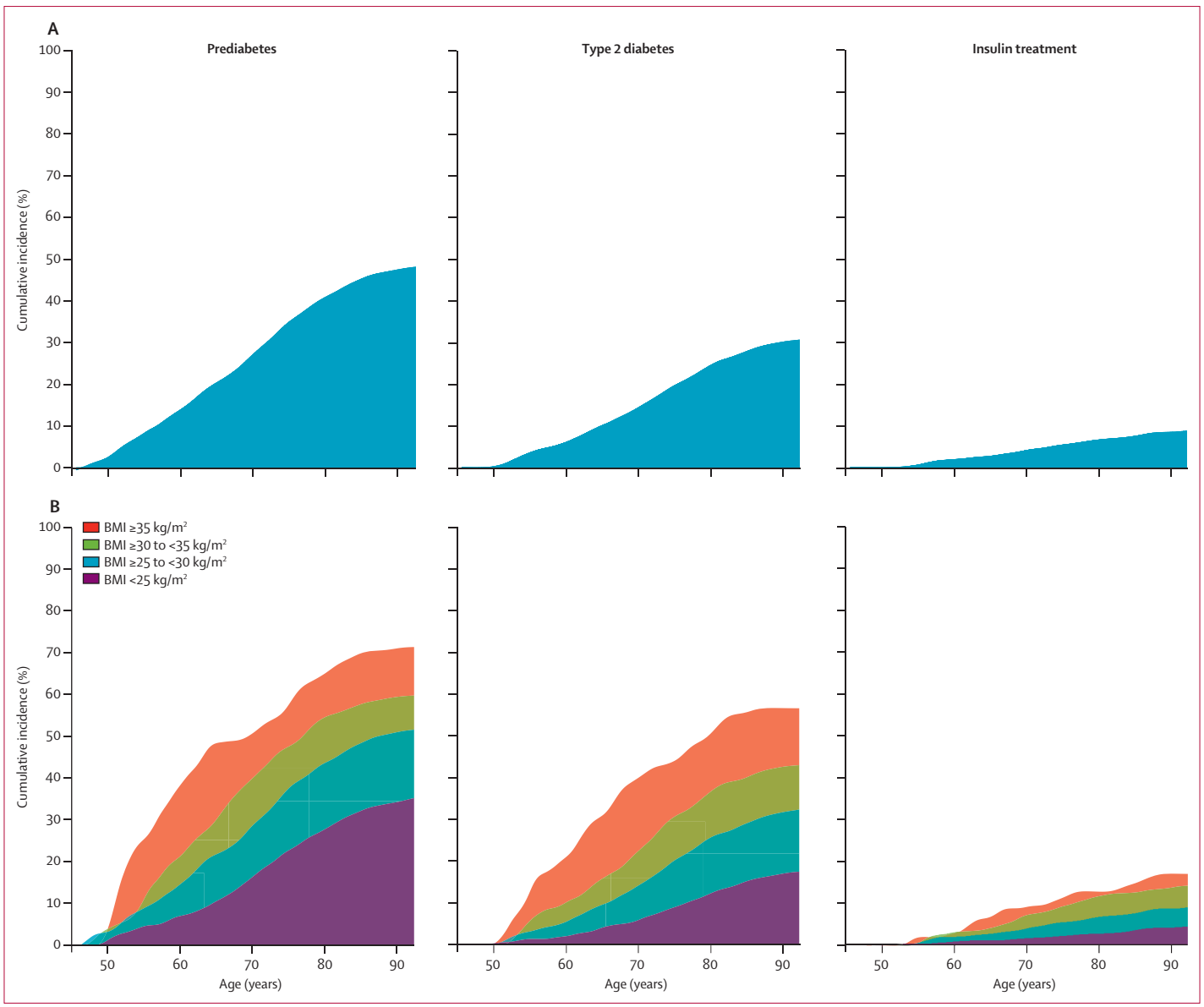


***Figure 2:*** **Lifetime risk of prediabetes, type 2 diabetes, and use of insulin treatment in individuals aged 45 years, adjusted for the competing risk of death**
(A) Cumulative incidence of prediabetes (fasting glucose >6·0 mmol/L), type 2 diabetes (fasting glucose ≥7·0 mmol/L or use of glucose-lowering drug), and insulin use in all individuals aged 45 years, adjusted for the competing risk of death. (B) Cumulative incidences of prediabetes, type 2 diabetes, and insulin use in individuals aged 45 years, adjusted for the competing risk of death, by BMI.

cumulative incidence by BMI strata as a function of age for individuals aged 45 years. When we stratified individuals within each BMI strata by waist circumference categories, we noted an increasing risk of diabetes with increasing waist size, except in the lowest BMI category (appendix p 10). The lifetime risk of progressing from prediabetes to diabetes was also substantially lower for individuals with a normal weight compared with overweight and obese individuals. However, the risk of starting insulin treatment in individuals with diabetes did not differ substantially between BMI strata (table 4).

Remaining life-years free of prediabetes, diabetes, and insulin use from age 45 years by sex, BMI, and waist circumference strata are shown in the appendix (appendix pp 5–6). Overall, years lived with normal glucose metabolism decreased with increasing levels of obesity. Additionally, years lived with diabetes increased with increasing BMI. For example, the average age of onset of prediabetes in men with normal weight was more than 10 years later compared with men with a BMI higher than 35 kg/m². On average, insulin treatment was only used for a short time at the end of life (appendix).

| | n | Lifetime risk of prediabetes (95% CI) | p value | n | Lifetime risk of diabetes (95% CI) | p value | n | Lifetime risk of insulin use (95%CI) | p value |
|---|---|---|---|---|---|---|---|---|---|
| **BMI (kg/m²)** | | | | | | | | | |
| <25 | 2686 | 36·9% (33·1–40·6) | .. | 2955 | 18·8% (15·8–21·7) | .. | 3124 | 4·6% (3·0–6·2) | .. |
| ≥25 to <30 | 3452 | 52·1% (48·0–56·2) | <0·0001 | 4132 | 33·1% (30·2–36·0) | <0·0001 | 4598 | 9·6% (7·7–11·5) | <0·0001 |
| ≥30 to <35 | 1013 | 60·2% (54·1–66·2) | 0·02 | 1324 | 43·9% (38·6–49·2) | 0·0002 | 1613 | 14·2% (10·5–17·9) | 0·01 |
| ≥35 | 240 | 71·3% (60·5–82·0) | 0·04 | 352 | 56·6% (46·7–66·6) | 0·01 | 462 | 17·1% (9·9–24·2) | 0·24 |
| **Waist circumference*** | | | | | | | | | |
| Small | 2127 | 37·5% (33·2–41·7) | .. | 2329 | 19·5% (16·0–23·1) | .. | 2449 | 4·5% (2·8–6·3) | .. |
| Medium | 2220 | 46·6% (41·1–52·2) | 0·005 | 2589 | 25·1% (21·6–28·6) | 0·01 | 2815 | 6·5% (4·4–8·5) | 0·08 |
| Large | 2801 | 57·8% (54·2–61·5) | 0·0005 | 3553 | 41·8% (38·5–45·0) | <0·0001 | 4205 | 12·6% (10·5–14·8) | <0·0001 |

Lifetime risk of prediabetes is calculated for individuals with normal glucose levels at the age of 45 years. Lifetime risk of diabetes is calculated for individuals without diabetes at the age of 45 years. Lifetime risk of insulin use is calculated for individuals not using insulin at the index age. *Waist circumference categories represent the WHO classification scheme (for men: small <94 cm, medium 94–102 cm, and large ≥102 cm; for women: <80 cm, 80–88 cm, and ≥88 cm). p values are for the comparison with the preceding BMI or waist category (eg, BMI ≥35 kg/m² with ≥30 to <35 kg/m²).

***Table 3:*** **Lifetime risk at age 45 years for prediabetes, diabetes, and insulin use, by BMI and waist circumference strata**

| | n | Prediabetes to diabetes (95% CI) | p value | n | Non-insulin-dependent diabetes to insulin use (95% CI) | p value |
|---|---|---|---|---|---|---|
| Unstratified | 1382 | 74·0% (67·6–80·5) | .. | 1043 | 49·1% (38·2–60·0) | .. |
| BMI (kg/m²) | | | | | | |
| <25 | 269 | 35·9% (18·4–53·4) | .. | 169 | 58·6% (34·2–83·1) | .. |
| ≥25 to <30 | 680 | 76·3% (68·9–83·7) | <0·0001 | 466 | 57·5% (45·2–69·9) | 0·47 |
| ≥30 to <35 | 311 | 87·7% (79·4–96·0) | 0·02 | 289 | 25·0% (0·0–60·1) | 0·04 |
| ≥35 | 112 | 80·9% (65·9–95·9) | 0·22 | 110 | 46·4% (28·6–64·1) | 0·14 |
| Waist circumference* | | | | | | |
| Small | 202 | 49·8% (28·6–70·9) | .. | 120 | 43·0% (24·4–61·7) | .. |
| Medium | 369 | 70·0% (56·4–83·7) | 0·06 | 226 | 55·9% (38·6–73·2) | 0·16 |
| Large | 752 | 78·8% (70·4–87·1) | 0·14 | 652 | 45·0% (29·0–61·0) | 0·18 |

*Waist circumference categories represent the WHO classification scheme (for men: small <94 cm, medium 94–102 cm, and large ≥102 cm; for women: <80 cm, 80–88 cm, and ≥88 cm). p values are for the comparison with the preceding BMI or waist category (eg, BMI ≥35 kg/m² with ≥30 to <35 kg/m²).

***Table 4:*** **Lifetime risk for the development of diabetes in individuals with prediabetes at age 45 years and lifetime risk for use of insulin in individuals aged 45 years with diabetes but not using insulin, stratified by BMI and waist circumference**

## Discussion

Our data suggest that the lifetime risk of developing prediabetes for a normoglycaemic individual aged 45 years is one in two, and one in three non-diabetic individuals aged 45 years will develop diabetes. Three-quarters of individuals with prediabetes at age 45 years will eventually progress to diabetes, and half of the patients with diabetes at the same age will start insulin treatment. Furthermore, obesity substantially affects the risk of progressing from prediabetes to diabetes and decreases the years lived with normal glucose metabolism.

Individuals with prediabetes have an increased risk of diabetes, cardiovascular disease, cancer, and mortality.[20–22] Despite the high prevalence of prediabetes, estimates of what proportion of the normoglycaemic population will eventually present with prediabetes have not been previously published. There is increasing evidence for the preventive effects of both lifestyle and pharmacological interventions on the progression of prediabetes to diabetes.[23–25] Lifetime risk estimates might specify the proportion of individuals for whom early intervention would be applicable. Based on our results, half of the normoglycaemic population will eventually present with prediabetes and might qualify for potential interventions during their lifespan.

By contrast with diabetes, prediabetes is a more fluctuating health state. The lifetime risk estimates of prediabetes in our study included individuals who had at least one measurement of serum glucose concentration in the prediabetes range. However, individuals diagnosed with prediabetes could return to normoglycaemia. For example, in the Diabetes Prevention Program (DPP), 19% of the placebo group returned to normoglycaemia within 10 years.[26] In a study to assess pioglitazone for diabetes prevention, 65 (28%) of 228 patients returned to normoglycaemia in the placebo group during a median follow-up of 2·4 years. However, these estimates are based on a short time and longer follow-up might result in different estimates. Our lifetime estimates show that three in four individuals with a glucose concentration in

the prediabetes range at age 45 years progress to diabetes. These estimates provide a better long-term perspective for individuals who ever meet prediabetes criteria, irrespective of whether an individual remains prediabetic or returns to normoglycaemia in subsequent years. This finding is in agreement with the American Diabetes Association (ADA) expert panel, which suggested that 70% of the individuals with prediabetes progress to diabetes.[2] The higher prevalence of obesity in the USA raises the concern of even higher progression rates compared with the estimates from our European population.

A previous report[11] simulated the lifetime risk of diabetes in a US population based on questionnaire data for the adjudication of diabetes which does not comprise undiagnosed diabetes. Because the prevalence of undiagnosed diabetes is more than 25% in the USA,[27] the risk estimates in the US study are probably underestimated. Furthermore, in an Australian study[12] investigators estimated the lifetime risk of diabetes (31% at age 45 years) from two cross-sectional examinations (diabetes defined on the basis of fasting plasma glucose [≥7·0 mmol/L] and 2 h plasma glucose [≥11·1 mmol/L]) with a short time interval (5 years). However, the population studied had many withdrawals (39% [4251/10 788]) and the study did not have an active follow-up. In our analysis, we used active follow-up data and fasting glucose measurements as an objective and comprehensive assessment of diabetes diagnosis, which allowed us to provide accurate estimates of the entire range of impaired glucose metabolism.

We noted a substantial effect of obesity on the remaining lifetime risk of prediabetes, diabetes, and insulin use, which is in line with results of a previous report.[28] Additionally, obesity increased the risk of progressing from prediabetes to diabetes. This result is in agreement with the finding in the placebo group of the DPP study, in which obese individuals had a higher risk of progressing to diabetes (nine *vs* 14 cases per 100 person-years).[23] Furthermore, obesity reduced the time lived with normal glucose metabolism and affected the time lived within each glycaemic state, emphasising the importance of weight management in the prevention of impaired glucose metabolism.

We estimate that one in two patients with diabetes will eventually start insulin treatment. Together, the lifetime risks of diabetes and insulin use suggest a high burden of pharmacological treatment of diabetes in our European population. Despite the consensus statements from the ADA and the European Association for the Study of Diabetes in 2006[29] and 2009,[30] physicians take into consideration several non-standardised factors when initiating glucose-lowering treatment.[31] Therefore, our lifetime risks of insulin use might not reflect country-specific and population-specific prescription behaviours. Furthermore, recent developments in diabetes care include the initiation of insulin treatment for β-cell preservation, which was not common practice during our study period. Therefore, lifetime risk of 9·1% in non-insulin users at age 45 years might be an underestimate of the risk of insulin use in the context of present diabetes clinical care.

The strength of our study is the comprehensive assessment of the diagnosis of incident diabetes by use of a blood glucose-lowering treatment (identified from medical records from hospitals and general practitioners), standardised blood glucose measurements at the repeated study centre visits, and electronic linkage with pharmacy dispensing records in the study area. Additionally, we used data from a prospective population-based cohort study with long-term follow-up and adjusted the lifetime risks for the competing risk of death to avoid overestimation. The study also had some limitations. First, we calculated remaining lifetime risks at age 45 years because we did not have data for individuals younger than 45 years. Nevertheless, the cumulative incidence of type 2 diabetes before this age is low.[10,11] Furthermore, to estimate the lifetime risk in BMI and waist circumference strata, we used anthropometric data in participants older than 45 years because not all individuals entered the study at age 45 years. This method could have led to the misclassification of individuals across the different categories because BMI and waist circumference could have changed with age. Third, we used data from a completely unselected sample of the general Dutch population with high participation rates.[13] However, all studies requiring active participation are to some extent subject to the so-called healthy volunteer effect, which generally leads to slight underestimations of absolute risk estimates at short-term follow-up. However, this underestimation is reduced at long-term follow-up.[32] Finally, most of the Rotterdam Study participants are white and we therefore present lifetime risks for individuals from European ancestry.

#### Contributors

SL and TTWvH collected data, did data analyses, and contributed equally as lead authors. MJGL was involved in data analysis and writing of the report. MK took part in the study design and writing of the report. AH and BHCS were principal investigators who contributed to study design, data collection, and writing of the report. MvH was involved in data collection and writing of the report. EJGS contributed to study design, data collection, and writing of the report. OHF was the local principal investigator who participated in the design and implementation of the study and the writing of the report. AD was responsible for the overall supervision, and contributed to data analyses and writing of the report. All authors had access to the data, commented on the report drafts, and approved the final submitted version.

#### Declaration of interests

OHF works in ErasmusAGE, a centre for ageing research funded by Nestlé Nutrition (Nestec), Metagenics, and AXA. MJGL is supported by a Prins Bernhard Cultuurfonds Fellowship (30140588) and the De Drie Lichten Foundation (04/14). AD is supported by a grant from the Netherlands Organisation for Scientific Research (NWO; veni, 916.12.154), and an EUR Fellowship. The other authors declare no competing interests.

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